

Simulation Results

Smart-Root-Dumb-Tree Model

Search scope : Combat move selection only - attacks only 1 territory in a round

Selection policy : UCB1 with exploration constant $c = 1.414$

Expansion : Single node per iteration

Simulation policy : Uniform random rollout for all players - keeps the tree dumb and root smart - excludes purchase and placement delegates - the simulation move generation has no pruning apart from time budget limit

Forward model : Handles the logs from the game, provides the requested move for the MCTS player (tree selects for combat and random for other delegates)

State evaluation : Territory strength ration + depth based penalties for non-terminal states

Playout : Fast playout by speeing legal move genration using game heuristics. The combat move generator gives 1 move with cheapest unit for undefended territory and a set of winnable moves with [min, max] strength for defended territories

Time budget : 1s per combat turn

Max tree depth : 10

Goal : Isolate and measure combat decision quality independent of the other game delegates

The reachability check is perfomed in Cpp and the linked to the python script using Cython

Efficiency summary

MCTS Efficiency Summary (Game-wise)

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Game 1

Total rounds: 23

Iterations per round:

Mean: 25.17

Min : 22

Max : 41

Exploration_coverage (actions_explored / total_actions):

Mean: 0.3440

Min : 0.0004

Max : 1.0000

Avg Tree depth per round:

Mean: 9.7948

Min : 9.6250

Max : 10.0000

Game 2

Total rounds: 72
Iterations per round:
Mean: 26.29
Min : 15
Max : 46
Exploration_coverage (actions_explored / total_actions):
Mean: 0.2821
Min : 0.0000
Max : 1.0000
Avg Tree depth per round:
Mean: 9.8578
Min : 9.4667
Max : 10.0000
Game 3

Total rounds: 59
Iterations per round:
Mean: 21.59
Min : 8
Max : 54
Exploration_coverage (actions_explored / total_actions):
Mean: 0.1701
Min : 0.0000
Max : 1.0000
Avg Tree depth per round:
Mean: 9.8091
Min : 9.5714
Max : 10.0000
Game 4

Total rounds: 35
Iterations per round:
Mean: 26.77
Min : 23
Max : 48
Exploration_coverage (actions_explored / total_actions):
Mean: 0.4609
Min : 0.0018
Max : 1.0000
Avg Tree depth per round:
Mean: 9.8524
Min : 9.6400
Max : 10.0000
Game 5

Total rounds: 45
Iterations per round:
Mean: 25.71
Min : 21
Max : 41
Exploration_coverage (actions_explored / total_actions):
Mean: 0.2849
Min : 0.0000
Max : 1.0000
Avg Tree depth per round:

Mean: 9.8318

Min : 9.6250

Max : 10.0000

OVERALL AVERAGES (Across Games)

Iterations per round (game-averaged):

Mean: 25.11

Min: 8

Max: 54

Exploration coverage (game-averaged):

Mean: 0.31

Min: 0.00

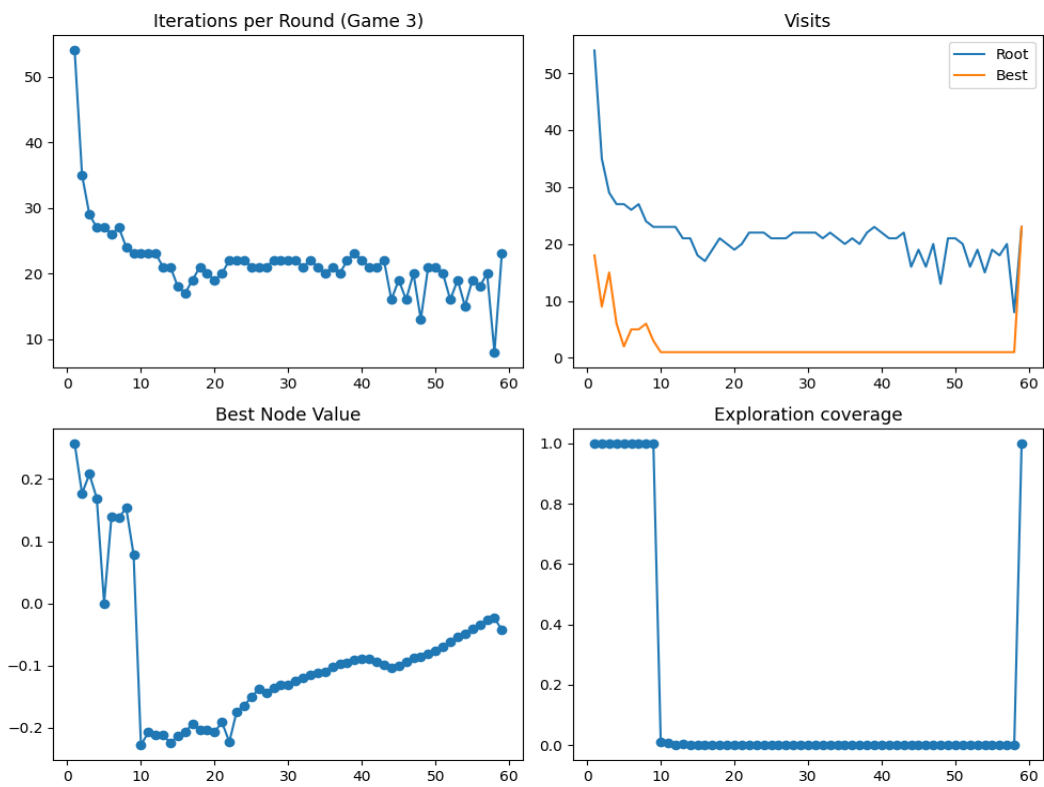
Max: 1.0

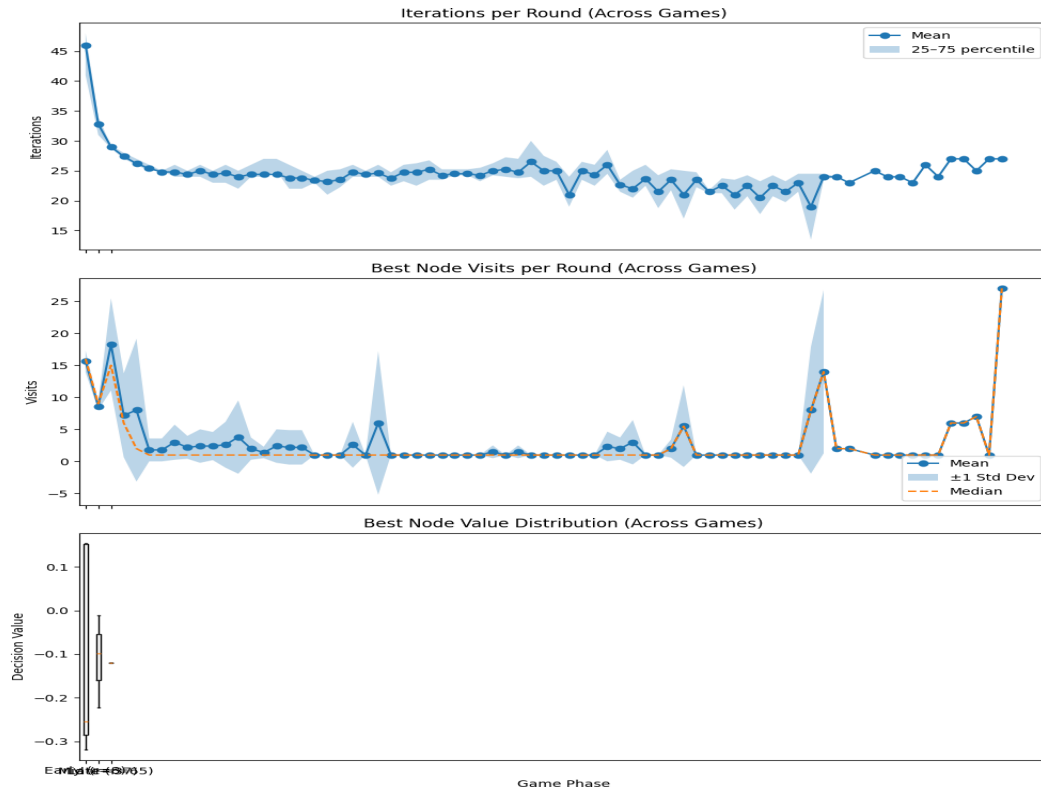
Avg tree depth per round (game-averaged):

Mean: 9.83

Min: 9.466666666666669

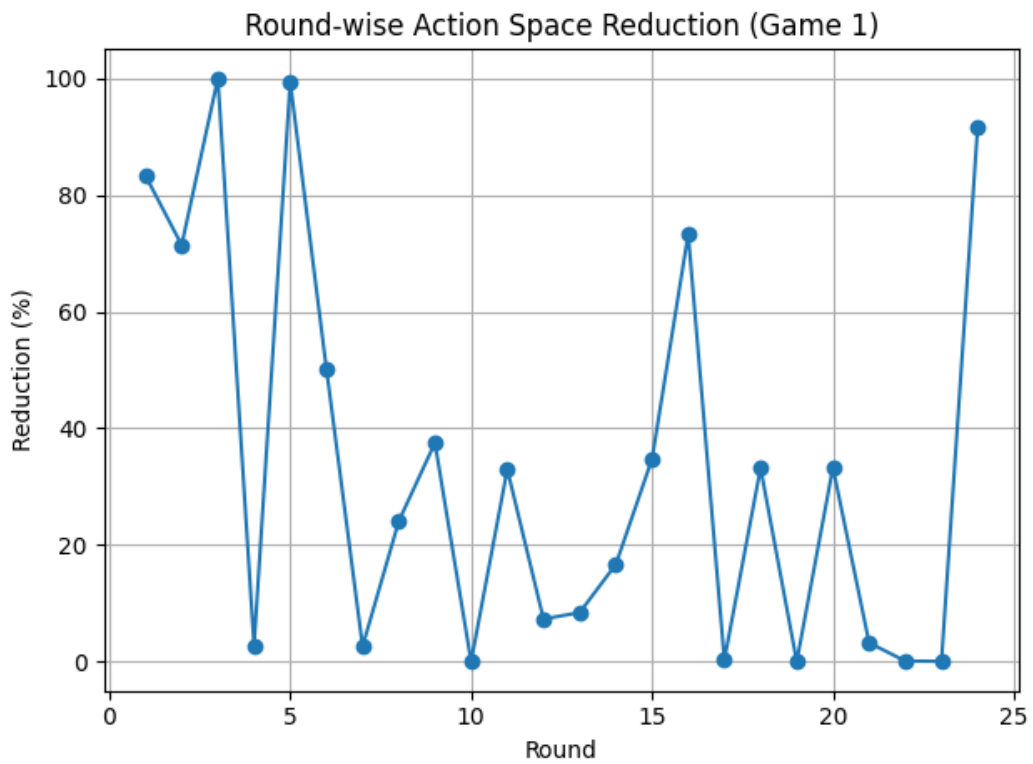
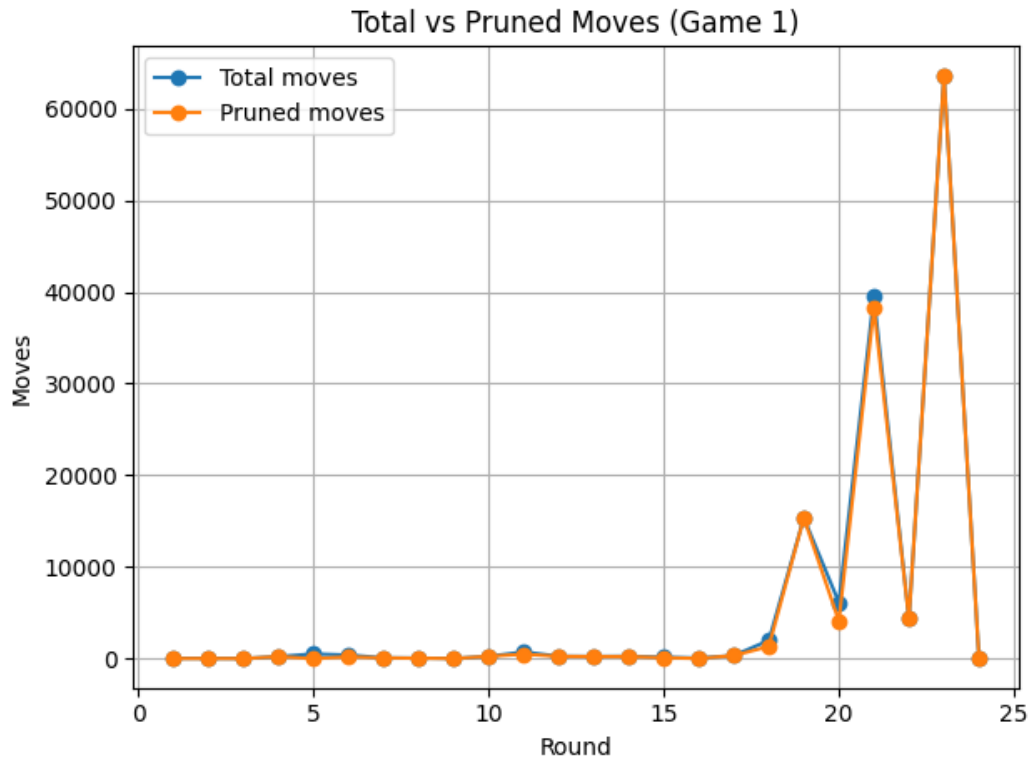
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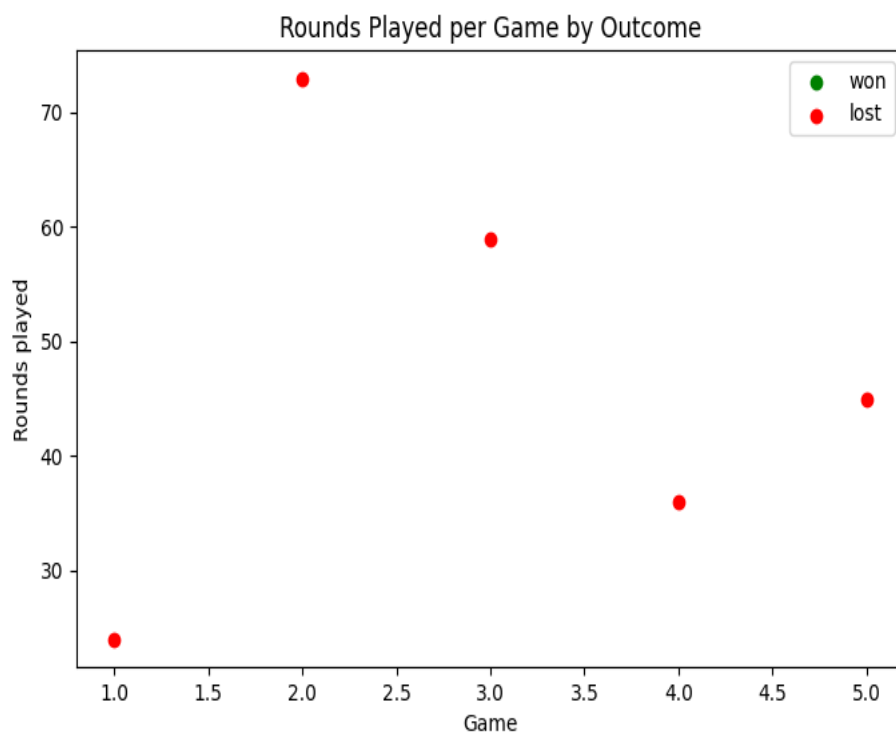
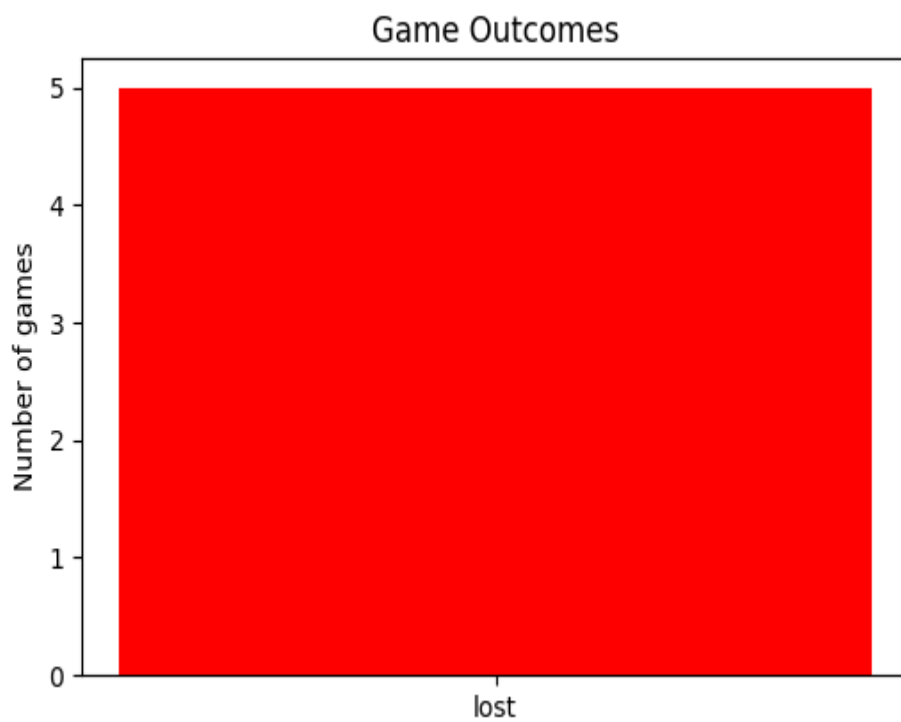


Action space reduction summary

game	rounds	avg_round_reduction	avg_round_reduction_percent
1	24.0	0.336059	33.605859
2	73.0	0.332324	33.232360
3	59.0	0.118250	11.825022
4	36.0	0.193684	19.368437
5	45.0	0.196695	19.669486
OVERALL	47.4	0.235402	23.540233



Game outcomes

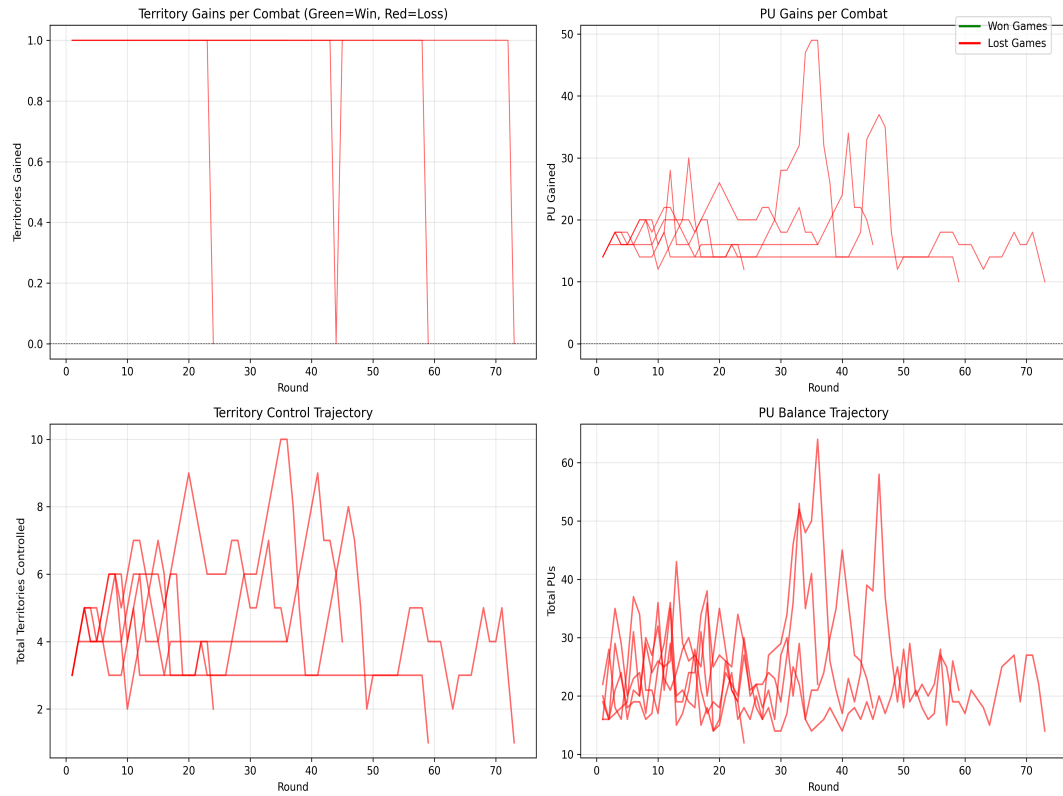


Combat move quality throughout the games

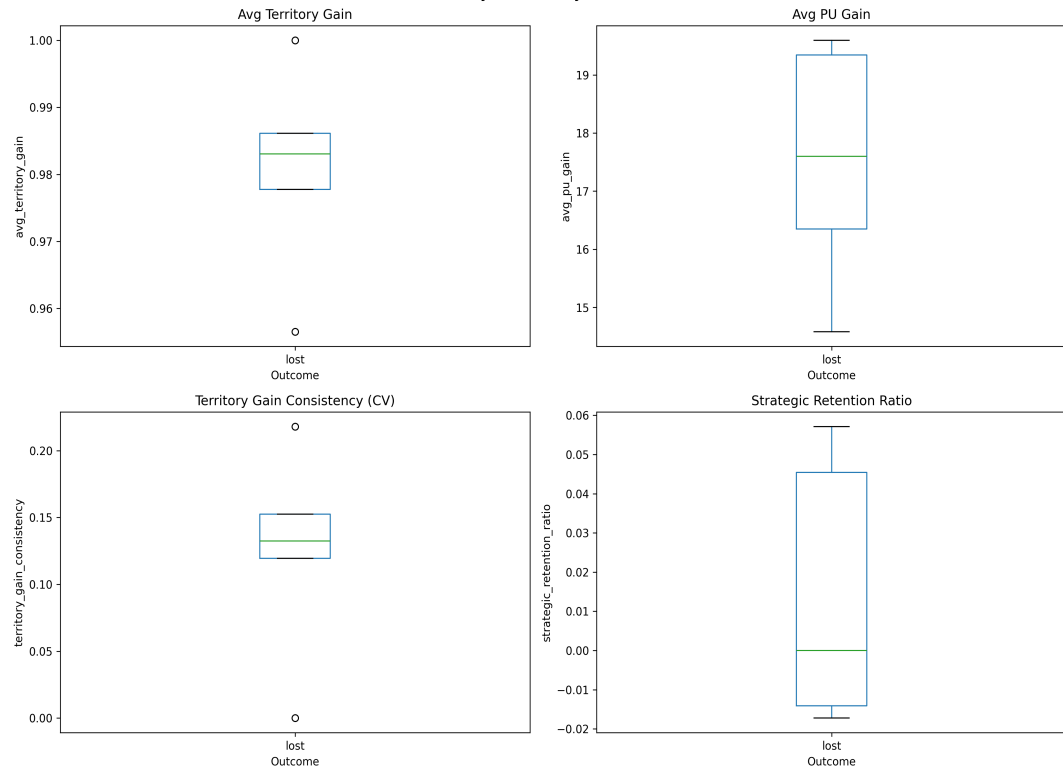
=== Combat Quality Metric Averages ===

Outcome: LOST

Avg Territory Gain: 0.981
Avg PU Gain: 17.494
Territory Gain Consistency: 0.124
Strategic Retention Ratio: 0.014



Combat Quality Metrics by Game Outcome



Profiling of MCTS algorithm

